

1. Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

```
Ethernet adapter VMware Network Adapter VMnet1:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::73fa:4d5d:eea7:c692%15
    IPv4 Address. . . . . : 192.168.73.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Ethernet adapter VMware Network Adapter VMnet8:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::ffda:c138:e611:987f%21
    IPv4 Address. . . . . : 192.168.17.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2405:201:2033:6025:95b1:f216:5d8c:9059
    Temporary IPv6 Address. . . . . : 2405:201:2033:6025:6042:d502:ff74:ddd
    Link-local IPv6 Address . . . . . : fe80::eec0:9248:fb84:eee8%13
    IPv4 Address. . . . . : 192.168.29.56
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::6ec2:42ff:fe23:edc9%13
                                192.168.29.1
```

2. Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

```
C:\Users\hellO>ping www.google.com

Pinging www.google.com [2001:4860:482c:7700::] with 32 bytes of data:
Reply from 2001:4860:482c:7700::: time=22ms
Reply from 2001:4860:482c:7700::: time=20ms
Reply from 2001:4860:482c:7700::: time=22ms
Reply from 2001:4860:482c:7700::: time=24ms

Ping statistics for 2001:4860:482c:7700:::
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 24ms, Average = 22ms
```

3. Run the **tracert** (or **traceroute** on Linux/Mac) command to **www.flipkart.com** and list the number of hops it takes to reach the destination.

```
C:\Users\hell0>tracert www.flipkart.com

Tracing route to flipkart.com [163.53.76.86]
over a maximum of 30 hops:

  1      5 ms      *          3 ms  reliance.reliance [192.168.29.1]
  2      6 ms     10 ms     23 ms  10.229.176.1
  3      6 ms      8 ms      6 ms  172.31.3.216
  4      8 ms     10 ms      7 ms  192.168.103.184
  5     52 ms      9 ms      4 ms  172.26.100.229
  6      6 ms     11 ms      6 ms  172.26.100.211
  7      8 ms      7 ms      4 ms  192.168.188.48
  8      *        *        *      Request timed out.
  9      *        *        *      Request timed out.
 10     67 ms     41 ms     30 ms  115.247.102.134
 11      *        *        *      Request timed out.
 12      *        *        *      Request timed out.
 13      *        *        *      Request timed out.
 14      *        *        *      Request timed out.
 15     30 ms     29 ms     30 ms  163.53.76.86

Trace complete.
```

4. Use the **nslookup** tool to find the IP address of **www.spotify.com** and explain in one line what this tool helps you verify.

```
C:\Users\hell0>nslookup www.spotify.com
Server:  reliance.reliance
Address:  2405:201:2033:6025::c0a8:1d01

Non-authoritative answer:
Name:     atc.spotify.map.fastly.net
Addresses: 2a04:4e42:31::810
           151.101.211.42
Aliases:  www.spotify.com
```

5. A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Hint: Think about checking both network connectivity and DNS resolution.

Step	Command	Purpose
1	ipconfig	Checks the computer's IP address, subnet mask, and default gateway to ensure the laptop has a valid network configuration.
2	ping zomato.com	Tests whether the laptop can reach Zomato's server and verifies basic network connectivity. It also checks for packet loss and response time.
3	nslookup zomato.com	Checks whether the domain name zomato.com is being correctly translated into an IP address by the DNS server. This helps identify DNS-related issues.

Brief Explanation

- **ipconfig** → Confirms the laptop has a valid IP configuration and gateway.
- **ping zomato.com** → Verifies that the network connection to the website is working.
- **nslookup zomato.com** → Confirms that DNS is resolving the website's domain name correctly.